Lecture 14: IV Standard Errors and Weak IV

In this lecture, we will discuss how to do inference for IV. First, we will note that IV is asymptotically normal and provide the appropriate standard error formula. From there however, we will note the issue of "weak instruments." When the first-stage coefficient is not "strong" (meaning we cannot confidently reject a first stage of zero), the typical asymptotic approximations work poorly: the distribution of $\hat{\beta}^{IV}$ will not be close to a normal distribution, and moreover it will be biased toward the OLS coefficient. We will discuss how to identify this issue, as well as a way to get confidence intervals that are valid even when we have a weak instrument problem.

Traditional Inference for IV

We will focus again on the case of a single instrument and a single regressor. Our regression model is:

$$Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$$

with the first-stage and reduced form:

$$X_i = \gamma + \pi \cdot Z_i + u_i$$
$$Y_i = \theta + \rho \cdot Z_i + \omega_i$$

Our IV estimator can then be expressed as:

$$\hat{\beta}^{IV} = \frac{\hat{\rho}}{\hat{\pi}} = \frac{\widehat{\text{Cov}}(Y_i, Z_i)}{\widehat{\text{Cov}}(X_i, Z_i)}$$

Plugging in the formula for Y_i, we get:

$$\begin{aligned}
\hat{\beta}^{IV} &= \frac{\widehat{\text{Cov}}(\alpha + \beta \cdot X_i + \varepsilon_i, Z_i)}{\widehat{\text{Cov}}(X_i, Z_i)} \\
&= \frac{\beta \cdot \widehat{\text{Cov}}(X_i, Z_i) + \widehat{\text{Cov}}(\varepsilon_i, Z_i)}{\widehat{\text{Cov}}(X_i, Z_i)} \\
&= \beta + \frac{\widehat{\text{Cov}}(\varepsilon_i, Z_i)}{\widehat{\text{Cov}}(X_i, Z_i)}
\end{aligned}$$

You may see a resemblance between this formula and the result we derived a few lectures ago for OLS: $\hat{\beta}^{OLS} = \beta + \frac{\widehat{\text{Cov}}(\varepsilon_i, X_i)}{\widehat{\text{Var}}(X_i)}$. This similarity is not coincidental: OLS is actually a special case of IV, where X_i is an instrument for itself!

The similar formula yields similar results: the IV estimator will be asymptotically normal, and we can derive a formula for the standard error. We will skip the details of the proofs, and instead just summarize the results.

First, under the same conditions that we used for consistency of IV, the IV estimator will be asymptotically normal. The theorem below formalizes this result and provides a formula for the asymptotic variance of IV:

Theorem (IV is Asymptotically Normal)**.** *Suppose that each observation is sampled iid, the relevance/full rank condition is satisfied, and the orthogonality condition holds. Then, the IV estimator (both simple IV and two-stage least squares) is asymptotically normal:* $\sqrt{N}\left(\hat{\beta}^{IV} - \beta\right) \xrightarrow{d} N(0, V)$*. Moreover, for the case of a single regressor and single instrument,* $V = \frac{Var((Z_i - \mathbb{E}[Z])\varepsilon_i)}{Cov(X_i, Z_i)^2}$.

Aside from the different formula for the standard error, inference for IV works the same as it does for OLS:

- We can use the plug-in-principle to get a robust, consistent standard error for IV:

$$\hat{V} = \frac{\widehat{\text{Var}}\left(\left(Z_i - \bar{Z}\right)\hat{\varepsilon}_i\right)}{\widehat{\text{Cov}}\left(X_i, Z_i\right)^2}$$
$$\hat{V} \xrightarrow{p} V$$
$$\widehat{\text{SE}}\left(\hat{\beta}^{IV}\right) = \sqrt{\frac{\hat{V}}{N}} = \sqrt{\frac{1}{N} \cdot \frac{\widehat{\text{Var}}\left(\left(Z_i - \bar{Z}\right)\hat{\varepsilon}_i\right)}{\widehat{\text{Cov}}\left(X_i, Z_i\right)^2}}$$

- The t-statistic will be asymptotically standard normal under the null:

$$\frac{\hat{\beta}^{IV} - \beta}{\widehat{\text{SE}}\left(\hat{\beta}^{IV}\right)} \xrightarrow{d} N\left(0, 1\right)$$

  This allows us to do hypothesis testing as usual, and also lets us construct confidence intervals:

$$\text{Confidence Interval: } \hat{\beta}^{IV} \pm c \cdot \widehat{\text{SE}}\left(\hat{\beta}^{IV}\right)$$

- Linear hypothesis testing with multiple coefficients and/or multiple equations also work the same way that they did for OLS. You can still use the `lincom` and/or `test` commands after running `ivregress`.
- Always use robust standard errors!

Comparing the IV and OLS Standard Errors

Which will have a smaller standard error, least squares or instrumental variables? It turns out that in the homoskedastic case,[1] we can show that IV will always have a standard error at least as large as OLS. For the IV standard error, we have:

$$\begin{aligned}\widehat{\text{SE}}_{\text{Homoskedastic}}\left(\hat{\beta}^{IV}\right) &= \sqrt{\frac{1}{N} \cdot \frac{\widehat{\text{Var}}\left(Z_i\right)\widehat{\text{Var}}\left(\hat{\varepsilon}_i^{IV}\right)}{\widehat{\text{Cov}}\left(X_i, Z_i\right)^2}} \\ &= \sqrt{\frac{1}{N} \cdot \frac{\widehat{\text{Var}}\left(\hat{\varepsilon}_i^{IV}\right)}{\underbrace{\widehat{\text{Corr}}\left(X_i, Z_i\right)^2}_{R^2 \text{ of First Stage}} \cdot \widehat{\text{Var}}\left(X_i\right)}}\end{aligned}$$

Which we can compare to the OLS homoskedastic standard error:

$$\widehat{\text{SE}}_{\text{Homoskedastic}}\left(\hat{\beta}^{OLS}\right) = \sqrt{\frac{1}{N} \cdot \frac{\widehat{\text{Var}}\left(\hat{\varepsilon}_i^{OLS}\right)}{\widehat{\text{Var}}\left(X_i\right)}}$$

The numerator, $\widehat{\text{Var}}\left(\hat{\varepsilon}_i\right)$, must be either smaller for OLS or the same: least squares is minimizing the sample variance of the residuals, so IV cannot do better than it. The denominator is $\widehat{\text{Var}}\left(X_i\right)$ for OLS, but is $\widehat{\text{Corr}}\left(X_i, Z_i\right)^2 \cdot \widehat{\text{Var}}\left(X_i\right)$ for IV, where we note that $\widehat{\text{Corr}}\left(X_i, Z_i\right)^2$ is also the R^2 of the first-stage regression. Since the R^2 must be non-negative and less than or equal to one, it must be that the denominator is either the same or larger for OLS than for IV.

This combines to mean that the (homoskedastic) OLS standard error will be smaller than the IV standard error. Although this is no longer a mathematical certainty in the heteroskedastic case, it is still almost always

[1] For OLS, homoskedasticity means $\mathbb{E}\left[\varepsilon_i^2 \mid X_i\right] = \mathbb{E}\left[\varepsilon_i^2\right]$. For IV, homoskedasticity instead means $\mathbb{E}\left[\varepsilon_i^2 \mid Z_i\right] = \mathbb{E}\left[\varepsilon_i^2\right]$.

true that the robust IV standard error is larger than the robust OLS standard error. Intuitively, IV is throwing out part of the variation in X_i, and in doing so it is throwing away statistical power.

This connects to a general idea in statistics: the bias-variance tradeoff. OLS is a more "efficient" estimator: it achieves a lower variance than IV. However, OLS is potentially biased, while IV will be consistent if the instrument is valid.

Testing for Endogeneity

Since OLS is more efficient than IV, we would prefer to use OLS if both methods are unbiased. We can test the hypothesis that OLS is biased by using a test of endogeneity. This tests the null hypothesis that $\beta^{OLS} = \beta^{IV}$. To implement this in Stata, we first run our IV regression (with robust standard errors), and then we run:

```
estat endogenous
```

This command will actually output two p-values, from two slightly different versions of this test. They are usually similar to each other, but if you want a reason to choose one over the other, the "robust score test" is outputted first, and is based on a test invented by Jeffrey Wooldridge, author of a popular econometrics textbook.

Note that although we call this a test of endogeneity, really we are only able to test whether $\beta^{OLS} = \beta^{IV}$. If the IV estimator is not consistent, then our test will not be comparing β^{OLS} to the true β.

Weak Instruments

For instrumental variables, the strength of the first-stage relationship between X_i and Z_i is crucial. Note that we can write the IV standard error as follows:

$$\begin{aligned}\widehat{\text{SE}}\left(\hat{\beta}^{IV}\right) &= \sqrt{\frac{1}{N} \cdot \frac{\widehat{\text{Var}}\left(\left(Z_i - \bar{Z}\right)\hat{\varepsilon}_i\right)}{\widehat{\text{Cov}}\left(X_i, Z_i\right)^2}} \\ &= \sqrt{\frac{1}{N} \cdot \frac{\widehat{\text{Var}}\left(\left(Z_i - \bar{Z}\right)\hat{\varepsilon}_i\right)}{\hat{\pi}^2 \cdot \widehat{\text{Var}}\left(Z_i\right)^2}}\end{aligned}$$

The standard error is inversely proportional not just to the variance of Z_i, but also to the first-stage coefficient π. If the first-stage coefficient is small, then that will inflate the standard error.

Unfortunately, large standard errors are the least of our problems with weak instruments. When the first-stage is weak, our standard asymptotics can break down.

Suppose that the first stage is close to zero, meaning that the first-stage coefficient, π, is small relative to the first-stage standard error, $\text{SE}\left(\hat{\pi}\right)$.[2] Recall that our IV estimator is the ratio of the reduced form and first stage: $\hat{\beta}^{IV} = \frac{\hat{\rho}}{\hat{\pi}}$. If $\text{SE}\left(\hat{\pi}\right)$ is large compared to π, then we will get two related problems.

First, we will often get a draw of $\hat{\pi}$ that is close to zero, or even has the wrong sign. This is like dividing by zero: if dividing by zero yields infinity, then dividing by a number close to zero yields very large numbers (which may be positive or negative). The result is that if the first stage is weak, then the distribution of $\hat{\beta}^{IV}$ will no longer be approximately normal. Instead, it will be a distribution with heavy tails: we will sometimes get draws of $\hat{\beta}^{IV}$ that are very large (positive or negative), and this will happen more often than the normal distribution would predict.

The second problem is that when the first-stage is weak, $\hat{\beta}^{IV}$ will be biased toward the OLS coefficient, β^{OLS}. It turns out that although the IV estimator is consistent, it has a finite-sample bias toward the OLS coefficient. This finite sample bias comes from estimation error in the first stage. In a large sample and with a strong first stage, the signal of the instrument will dominate the noise of estimation error. But if the instrument has little signal, then IV will be biased toward OLS.

[2] Notice here that I am referring to the true first-stage coefficient, π, and the true standard error of $\hat{\pi}$.

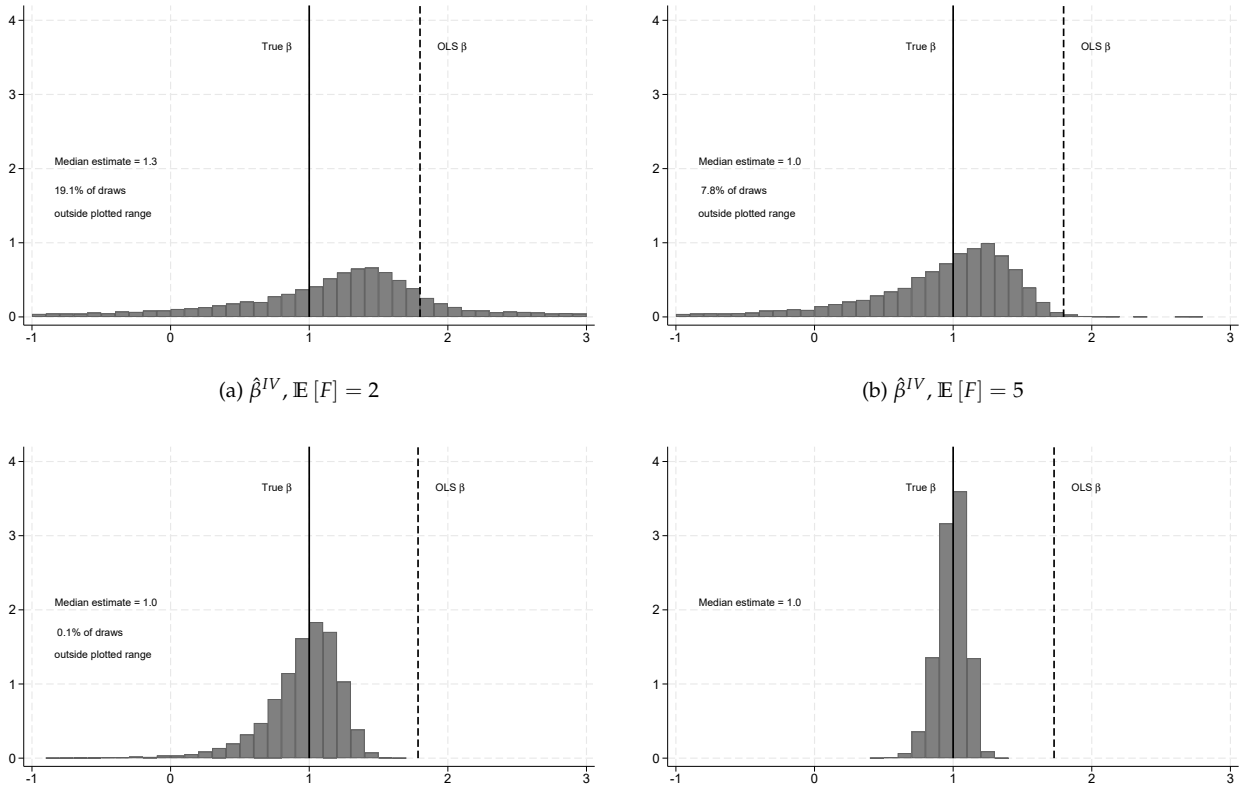
Simulations with Weak Instruments

To explore these issues, we will simulate the distribution of $\hat{\beta}^{IV}$ under varying instrument strength. In each simulation, we will have the main equation and the first-stage:

$$Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$$
$$X_i = \gamma + \pi \cdot Z_i + u_i$$

where Z_i and u_i are independent, standard normal random variables. The error term ε_i is also drawn from a standard normal distribution; it is correlated with u_i with Corr $(u_i, \varepsilon_i) = 0.8$, and uncorrelated with Z_i.[3] All variables are jointly normal. Each simulation has 1,000 observations, with each observation drawn i.i.d. The correlation between u_i and ε_i means that the OLS coefficient will be higher than the true coefficient, but IV will be consistent since Z_i is uncorrelated with ε_i.

We set the true β to 1, and we set α and γ to zero (the intercepts actually have no effect on the simulation, so this is innocuous). We vary the first-stage coefficient π across simulations to give us different values for $\mathbb{E}[F]$, where F is the F-statistic for the null hypothesis that $\pi = 0$. We draw 10,000 independent samples, and for each simulation we compute the IV estimator $\hat{\beta}^{IV}$and the t-statistic, $\left(\hat{\beta}^{IV} - 1\right) / \widehat{\text{SE}}\left(\hat{\beta}^{IV}\right)$, which tests the true null that $\beta = 1$.



(a) $\hat{\beta}^{IV}$, $\mathbb{E}[F] = 2$

(b) $\hat{\beta}^{IV}$, $\mathbb{E}[F] = 5$

(c) $\hat{\beta}^{IV}$, $\mathbb{E}[F] = 20$

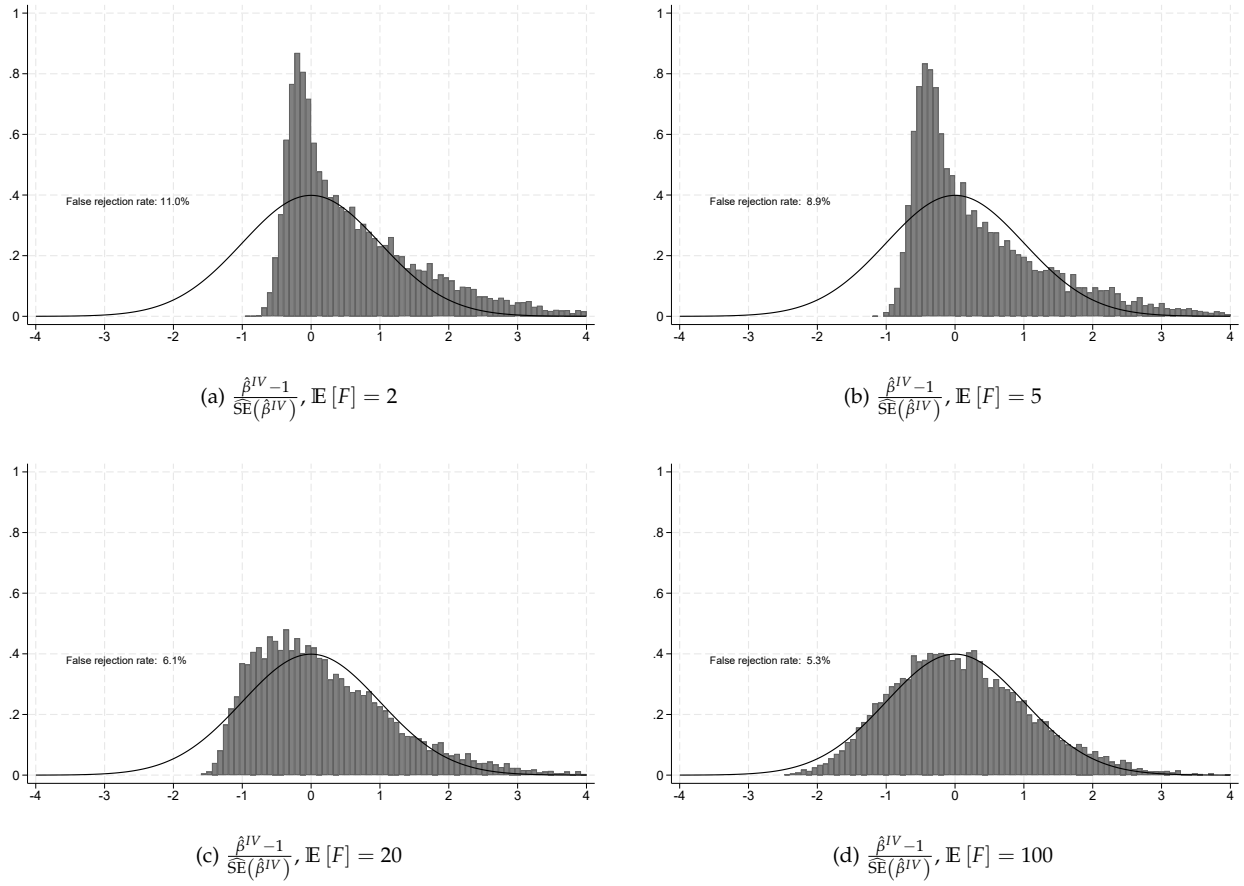
(d) $\hat{\beta}^{IV}$, $\mathbb{E}[F] = 100$

Figure 1: Distribution of Instrumental Variables Estimator, $\hat{\beta}^{IV}$, for Varying Instrument Strength

Figure 1 shows the simulated distribution of $\hat{\beta}^{IV}$. The top-left panel shows results for $\mathbb{E}[F] = 2$, reflecting a weak instrument. The sampling distribution is clearly not normal, with heavy tails (note that roughly one fifth of simulation draws are outside the range of the figure). Moreover, the median estimate is 1.3, reflecting bias toward the OLS coefficient. As we move across the panels, the instrument gets stronger and

[3] To implement this, we draw $\varepsilon_i = 0.8u_i + \sqrt{1 - 0.8^2}v_i$, where v_i is also an independently drawn standard normal random variable.

our IV estimator begins to behave more "normally:" the sampling distribution begins to look like a normal distribution. Moreover, as the instrument gets stronger the distribution becomes centered around the true coefficient. A stronger instrument also shrinks the standard error, and thus we see the distribution become more concentrated as $\mathbb{E}[F]$ increases.



(a) $\frac{\hat{\beta}^{IV}-1}{\widehat{\text{SE}}(\hat{\beta}^{IV})}$, $\mathbb{E}[F] = 2$

(b) $\frac{\hat{\beta}^{IV}-1}{\widehat{\text{SE}}(\hat{\beta}^{IV})}$, $\mathbb{E}[F] = 5$

(c) $\frac{\hat{\beta}^{IV}-1}{\widehat{\text{SE}}(\hat{\beta}^{IV})}$, $\mathbb{E}[F] = 20$

(d) $\frac{\hat{\beta}^{IV}-1}{\widehat{\text{SE}}(\hat{\beta}^{IV})}$, $\mathbb{E}[F] = 100$

Figure 2: Distribution of t-statistic for Varying Instrument Strength

Figure 2 shows the simulated distribution of the t-statistic testing the null $\beta = 1$. For comparison, we also plot the density of the standard normal distribution. Each panel also reports the probability that a hypothesis test would reject the null using this t-statistic and using a critical value of 1.96 based on the standard normal distribution (rejecting if the absolute value of the t-statistic is above the critical value). Since the null is true in the simulation, the hypothesis test is supposed to reject 5% of the time. Yet when the instrument is weak, the distribution of the t-statistic deviates dramatically from the standard normal distribution. As a result, we get substantial over-rejection: when $\mathbb{E}[F] = 2$, a "5% hypothesis test" actually rejects 11% of the time. The false rejection rate falls as the instrument gets stronger, and is roughly 5% when $\mathbb{E}[F] = 100$.

Testing for Weak Instruments

Weak instruments are a major problem with instrumental variables. It's thus important to test whether our instruments are strong. If we have one instrument and one regressor, then we can run the first-stage regression and compute the F-statistic for the null hypothesis that $\pi = 0$ (note that the F-statistic for a single equation is just the t-statistic squared). The standard rule of thumb is that if $F > 10$ then the instru-

ment is strong enough to proceed without worrying further about weak instruments.[4] If we have multiple instruments, the state-of-the-art test is the Montiel-Olea-Pflueger test, available through the `weakivtest` command.[5]

Weak-Instrument-Robust Confidence Intervals

If we find that our instrument is weak, not all hope is lost. There are alternative methods to construct confidence intervals that are robust to weak instruments. The most prominent of these methods is the Anderson-Rubin test, or AR test.

The Anderson-Rubin test works by directly testing the orthogonality condition. The orthogonality condition states that $\varepsilon_i = Y_i - \alpha - \beta \cdot X_i$ will be uncorrelated with the instrument. So, if $\beta = \beta_0$, this implies that $Y_i - \beta_0 \cdot X_i$ should be uncorrelated with the instrument (the intercept, α, can be excluded because it does not affect the covariance). Thus, we can run the following regression:

$$Y_i - \beta_0 \cdot X_i = \alpha + \delta \cdot Z_i + \varepsilon_i$$

where we have constructed the outcome variable as $Y_i - \beta_0 \cdot X_i$. If the null hypothesis is true, then $Y_i - \beta_0 \cdot X_i = \alpha + \varepsilon_i$, and thus δ will equal zero. We can then test our null hypothesis by testing the null $H_0 : \delta = 0$, and this will be a valid test of the null hypothesis $H_0 : \beta = \beta_0$. We can construct confidence intervals by running the test for each value of β_0, and keeping the values that fail to reject (in practice, we run this test for a grid of values). Note that the Anderson-Rubin test can take unusual shapes: they can have holes in them and/or extend out to infinity.

To implement this in Stata, we first run our IV regression (with robust standard errors), and then we run:

```
estat weak, ci
```

Note that these methods only work well for a single endogenous regressor: we still do not have good techniques to handle weak instruments and multiple endogenous regressors.

[4] Some econometricians argue that the threshold should be (much) higher than 10: the standard advice may be different in a few years!

[5] This command is not part of base Stata; to download the package run `ssc install weakivtest`.